```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;

class Car {
    private String carId;
    private String brand;
    private String model;
    private double basePricePerDay;
    private boolean isAvailable;

    public Car(String carId, String brand, String model, double basePricePerDay) {
        this.carId = carId;
        this.brand = brand;
        this.model = model;
        this.basePricePerDay = basePricePerDay;
        this.isAvailable = true;
    }
    public String getCarId() {
        return carId;
    }

    public String getBrand() {
        return brand;
    }

    public String getModel() {
        return model;
    }

    public double calculatePrice(int rentalDays) {
        return basePricePerDay * rentalDays;
    }

    public boolean isAvailable() {
       return isAvailable;
    }

    public void rent() {
        isAvailable = false;
    }

    public void returnCar() {
        isAvailable = true;
    }

}
class Customer {
    private String customerId;
    private String name;
```

```
    public Customer(String customerId, String name) {
        this.customerId = customerId;
        this.name = name;
    }

    public String getCustomerId() {
        return customerId;
    }

    public String getName() {
        return name;
    }
}

class Rental {
    private Car car;
    private Customer customer;
    private int days;

    public Rental(Car car, Customer customer, int days ) {
        this.car = car;
        this.customer = customer;
        this.days = days;
    }

    public Car getCar() {
        return car;
    }

    public Customer getCustomer() {
        return customer;
    }

    public int getDays() {
        return days;
    }
}

class CarRentalSystem {
    private List<Car>cars;
    private List<Customer>customers;
    private List<Rental>rentals;

    public CarRentalSystem() {
        cars = new ArrayList<>();
        customers = new ArrayList<>();
        rentals = new ArrayList<>();
    }
```

```
public void addCar(Car car) {
    cars.add(car);
}

public void addCustomer(Customer customer) {
    customers.add(customer);
}

public void rentCar(Car car, Customer customer, int days) {
    if (car.isAvailable()) {
        car.rent();
        rentals.add(new Rental(car,customer, days));
    } else {
        System.out.println("Car is not available for rent.");
    }
}


public void returnCar(Car car) {
    car.returnCar();
    Rental rentalToRemove = null;
    for (Rental rental : rentals) {
        if (rental.getCar() == car) {
            rentalToRemove = rental;
            break;
        }
    }
    if (rentalToRemove != null) {
        rentals.remove(rentalToRemove);
        System.out.println("Car returned successfully.");
    } else {
        System.out.println("Car was not rented.");
    }
}

public void menu() {
    Scanner scanner = new Scanner(System.in);

    while(true) {
        System.out.println("==== Car rental System =====");
        System.out.println("1. Rent a Car");
        System.out.println("2. Return a Car");
        System.out.println("3. Exit");
        System.out.print("Enter your choice: ");

        int choice = scanner.nextInt();
        scanner.nextLine(); // Consume newLine

        if (choice == 1) {
            System.out.println("\n== Rent a Car ==\n");
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                System.out.print("Enter your name: ");
                String customerName = scanner.nextLine();

                System.out.println("\nAvailable Cars:");
                for(Car car : cars) {
                    if (car.isAvailable()) {
                        System.out.println(car.getCarId() + " - " + car.getBrand()
+ " " + car.getModel());
                    }
                }

                System.out.print("\nEnter the car ID you want to rent: ");
                String carId = scanner.nextLine();

                System.out.print("Enter the number of days for rental: ");
                int rentalDays = scanner.nextInt();
                scanner.nextLine(); // Consume newline

                Customer newCustomer = new Customer( "CUS" + (customers.size() +
1), customerName);
                addCustomer(newCustomer);

                Car selectedCar = null;
                for (Car car : cars) {
                    if (car.getCarId().equals(carId) && car.isAvailable()) {
                        selectedCar = car;
                        break;
                    }
                }

                if(selectedCar != null) {
                    double totalPrice = selectedCar.calculatePrice(rentalDays);
                    System.out.println("\n== Rental Information ==\n");
                    System.out.println("Customer ID: " +
newCustomer.getCustomerId());
                    System.out.println("Customer Name: " + newCustomer.getName());
                    System.out.println("Car: " + selectedCar.getBrand() + " " +
selectedCar.getModel());
                    System.out.println("Rental Days: " +rentalDays);
                    System.out.printf("Total Price: $%.2f%n", totalPrice);

                    System.out.print("\nConfirm rental(Y/N): ");
                    String confirm = scanner.nextLine();

                    if(confirm.equalsIgnoreCase( "Y")) {
                        rentCar(selectedCar, newCustomer,rentalDays);
                        System.out.println("\nCar rented successfully.");
                    } else {
                        System.out.println("\nRental canceled.");
                    }
```

```
                } else {
                    System.out.println("\nInvalid car selection or car not
available for rent.");
                }
            } else if (choice == 2) {
                System.out.println("\n== Return a Car ==\n");
                System.out.print("Enter the car ID you want to return: ");
                String carId = scanner.nextLine();

                Car carToReturn = null;
                for (Car car : cars) {
                    if(car.getCarId().equals(carId) && !car.isAvailable()) {
                        carToReturn = car;
                        break;
                    }
                }

                if(carToReturn != null) {
                    Customer customer = null;
                    for(Rental rental : rentals) {
                        if(rental.getCar() == carToReturn) {
                            customer = rental.getCustomer();
                            break;
                        }
                    }

                    if (customer != null) {
                        returnCar(carToReturn);
                        System.out.println("Car rented successfully by " +
customer.getName());
                    }  else {
                        System.out.println("Car was not rented or rental
information is missing.");
                    }
                } else {
                    System.out.println("Invalid car ID or car is not rented.");
                }
            } else if(choice == 3) {
               break;
            } else {
                System.out.println("Invalid choice. Please enter a valid
option.");
            }
        }

        System.out.println("\nThank you for using the Car Rental System!");
    }
}

public class Main{
```

```
    public static void main(String[] args) {
        CarRentalSystem rentalSystem = new CarRentalSystem();

        Car car1 = new Car( "C001",  "Toyota",  "Canry",  60.0); // Different base
price per day for each car
        Car car2 = new Car( "C002", "Honda", "Accord",  70.0);
        Car car3 = new Car("C003",  "Mahindra",  "Thar",  150.0);
        rentalSystem.addCar(car1);
        rentalSystem.addCar(car2);
        rentalSystem.addCar(car3);

        rentalSystem.menu();
    }
}
```